The Real Effects of Short-Selling Constraints

Gustavo Grullon
Rice University

Sébastien Michenaud
DePaul University

James P. Weston
Rice University

We use a regulatory experiment (Regulation SHO) that relaxes short-selling constraints on a random sample of U.S. stocks to test whether capital market frictions have an effect on stock prices and corporate decisions. We find that an increase in short-selling activity causes prices to fall, and that small firms react to these lower prices by reducing equity issues and investment. These results not only provide evidence that short-selling constraints affect asset prices, but also confirm that short-selling activity has a causal impact on financing and investment decisions. (*JEL* G18, G31, G32)

Can capital market frictions affect stock prices and real economic activity? In this paper, we use a natural experiment to test the hypothesis that short-selling constraints have a causal effect on stock prices and corporate behavior. Our experiment centers on a regulatory change (Regulation SHO) that caused an increase in short-selling activity. Because this regulation affects a randomly selected treatment group, its impact on stock prices and real corporate activities is exogenous to investment opportunities. We find that the stock prices of firms in the treatment group fall relative to the control group. More importantly, this exogenous change in prices leads to a change in corporate policy, especially for small firms.

There are two main channels through which short-selling constraints can affect both stock prices and real corporate decisions. Studies by Miller (1977),

We thank Nina Baranchuk, Paul Bennett, Alex Butler, François Degeorge, François Derrien, James Dow, Laurent Frésard, Vincent Glode, Itay Goldstein (the editor), Ulrich Hege, Johan Hombert, Jiekun Huang, Yamil Kaba, Ambrus Kecskés, Paul Koch, Holger Spamann, Phillip Valta, Nadia Vozlyublennaia, Morad Zekhnini, two anonymous referees, and seminar participants at the 2012 European Finance Association Meetings, the 8th Penn/NYU Conference on Law and Finance, 2012 Financial Management Association Meetings, Fordham University, Rice University, University of Alabama, University of Kansas, University of Texas at Austin, Pennsylvania State University, Singapore Management University, National University of Singapore, Nanyang Technological University, Hong Kong Science and Technology University, and the University of Western Ontario for useful comments and suggestions. All remaining errors are our own. Michenaud was affiliated with Rice University when this work was conducted. Send correspondence to Gustavo Grullon, Rice University, 6100 Main St, Houston TX 77005; telephone: (713) 3486138. E-mail: grullon@rice.edu.

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doi:10.1093/rfs/hhv013 Advance Access publication February 12, 2015

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Harrison and Kreps (1978), Allen, Morris, and Postlewaite (1993), Chen, Hong, and Stein (2002), Scheinkman and Xiong (2003), and Hong and Stein (2003) all demonstrate that short-selling constraints can lead to overvaluation by making it harder for prices to reflect negative information. Gilchrist, Himmelberg, and Huberman (2005) show that this overvaluation can cause overinvestment by artificially reducing the firm's cost of capital. All this suggests that, in the presence of overvaluation, removing short-selling constraints should lead to lower stock prices and investment.

In addition to the overvaluation channel, recent theoretical work suggests that short-selling constraints can have real effects by mitigating the impact of strategic traders (bear raiders). For example, in Goldstein and Guembel (2008), an exogenous change in prices can lead to cancellation of value-creating investment projects because managers learn from stock prices to make real investment decisions. As a result, bear raiders have an incentive to manipulate stock prices to distort firms' investment policy.[1] Goldstein and Guembel (2008) show that this feedback effect from the stock market to firms' real decisions incentivizes bear raiders to short stocks even when these traders are uninformed.

Empirically, the evidence on the effects of short selling on stock prices is mixed. While there is some evidence that short-selling restrictions lead to stock overvaluation (e.g., Jones and Lamont 2002; Ofek and Richardson 2003; Ofek, Richardson, and Whitelaw 2004; and Cohen, Diether, and Malloy 2007), other studies find that short-selling constraints have only a negligible effect on prices (e.g., Battalio and Shultz 2006; Diether, Lee, and Werner 2009; Beber and Pagano 2013; and Kaplan, Moskowitz, and Sensoy 2013).[2] Further, there is little empirical evidence linking short-selling constraints to corporate decisions. Of course, a primary challenge in studying this new question is identifying causal relationships when firm fundamentals, short selling, and prices are often jointly determined.

In this paper we use an identification strategy based on the SEC's 2004 approval of Regulation SHO, which announced the removal of restrictions on short sales in 2005. Since the 1930s, short sales could not be placed when stock prices were declining, a regulation commonly referred to as the uptick rule.[3] However, the SEC lifted this restriction in 2005 for a randomly selected sample of one-third of the Russell 3000 stocks (the pilot group). Two years later the SEC removed the restriction for all stocks after analyzing the results from the experiment.

[1] Subrahmanyam and Titman (2001), Khanna and Sonti (2004), and Goldstein, Ozdenoren, and Yuan (2013), among others, examine alternative mechanisms through which stock prices can affect firms' fundamentals.

[2] Finding that short-selling constraints do not affect the level of stock prices is consistent with the predictions in Diamond and Verrechia (1987) and Gallmeyer and Hollifield (2008). This effect could be due to the increase in the speed of price adjustment to information arrival (see, for example, Arnold, Butler, Crack, and Zhang 2004).

[3] Rule 10a-1 imposed the uptick rule on the NYSE in 1938, and Rule 3350 imposed the bid price test on NASDAQ in 1994. On the NYSE, short sales can only be made on plus ticks or zero plus ticks based on the last sale. The bid test applicable to NASDAQ securities prohibits sales below the bid if the last bid was a down bid.

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The purpose of the uptick rule, in place since 1938, was to limit short-selling activity.[4] Consistent with this regulatory goal, previous studies find that the uptick rule significantly impedes the execution of short sales (see, e.g., Angel 1997 and Alexander and Peterson 1999). Chung (1991) finds that the uptick rule can be an important constraint to index arbitrage. Further, Lamont (2012) provides evidence of firms deliberately switching exchanges to get protection from the uptick rule. In this context, the removal of the short-sale constraints may cause an increase in regulatory uncertainty for firms, which could have economically significant effects on stock prices, investment, and security issuance.

Not surprisingly, Regulation SHO was not welcomed by the exchanges or listed firms. In public comments, NYSE officials, specialists, and member firms all expressed support for short-sale restrictions and opposed any change that affected some listed stocks but not others.[5] Further, in a 2008 NYSE survey, 85% of CEOs, CFOs, and investor relation officers surveyed were in favor of reinstituting the uptick rule as soon as possible. In general, managers appear sensitive to the impact of short sellers on their stock prices.

While such fears have theoretical support, studies examining the effect of Reg SHO on trading behavior and prices find a positive effect on short-selling activity and stock market quality but no effect on prices or volatility (Diether, Lee, and Werner 2009; Alexander and Peterson 2008; Boehmer, Jones, and Zhang 2008; Office of Economic Analysis [OEA] 2007). These studies, however, focus on a short window around both the announcement and the effective date of this regulatory change.

We reexamine these findings using a wider event window and find different results. We find evidence of abnormal stock returns over different time horizons, but the initial effect on prices happens about two weeks before the SEC announcement date (but after the SEC board approval date), suggesting that information was incorporated into prices prior to the public announcement. Two weeks prior to the announcement, the cumulative abnormal return is -1.52% for all the firms in the pilot group and -2.35% for small firms. Moreover, small pilot firms underperform by nearly 900 basis points over the next two years. Short-selling activity also increases more for firms in the pilot group. Our findings are consistent with Allen, Morris, and Postlewaite (1993), who predict that bubbles may disappear when information creates common knowledge that

[4] The objectives of the uptick rule as stated by the SEC are: "allowing relatively unrestricted short selling in an advancing market; preventing short selling at successively lower prices, thus eliminating short selling as a tool for driving the market down; and preventing short sellers from accelerating a declining market by exhausting all remaining bids at one price level, causing successively lower prices to be established by long sellers."

[5] For example, see the March 1, 2004, open letter to the SEC by Darla C. Stuckey, Corporate Secretary of the New York Stock Exchange, and the February 12, 2004, open letter to the SEC by David Humphreville, President of the Specialist Association. The latter states that firms "fear, legitimately, that removal of Rule 10a-1's price constraints on short selling of their companies' stocks for two years will undermine proper pricing, tend to discourage new decisions to invest, and weaken the resolve of current holders of their companies' stocks to refrain from selling them in times of market stress."

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prices will fall in the future, even before short-selling constraints are removed. Further, they are consistent with the idea that removing short-selling constraints increases the probability of coordinated short-selling attacks (Goldstein and Guembel 2008).

Our results are consistent with recent papers that identify an effect of short selling on stock prices (Jones and Lamont 2002; Ofek and Richardson 2003; Ofek, Richardson, and Whitelaw 2004; Cohen, Diether, and Malloy 2007). However, our results contrast with those in Kaplan, Moskowitz, and Sensoy (2013), who investigate the effect of an exogenous shock to the supply of lendable shares and find little effect on asset prices. We believe our results are different because we use a larger scale experiment that was publicly announced, and removed a regulatory constraint that was in place for over 70 years.

Given the effect on prices, we test whether the exogenous shock to short selling affects firms' corporate decisions. While there is empirical evidence that equity prices may drive real investment decisions (see Chirinko and Schaller 2001; Goyal and Yamada 2004; Campello and Graham 2013; Baker, Stein, and Wurgler 2003; Polk and Sapienza 2009; Edmans, Goldstein, and Jiang 2012; Campello, Ribas, and Wang 2014; Hau and Lai 2013), there is no evidence relating the price impact of short-selling activity on investment. Since we identify an exogenous change in short-selling constraints, our natural experiment is well suited for examining this issue.

Using a difference-in-differences approach, we find that small constrained firms in the pilot program reduce capital expenditures by 8% to 11% relative to the control group. While large in percentage terms, the real effects need to be compared against a drop in stock prices of close to 9% over two years, or a \$53 million drop in market capitalization for the small firms in the sample. This decline of \$53 million in market capitalization resulted in a \$9.4 million drop in equity issues, and in a \$1.8 million drop in capital expenditures.

Next, we dig deeper into the cross-sectional variation of our results. Small financially constrained firms are harder to short and thus more likely to be overvalued. In addition, financially constrained firms are more susceptible to bear raids (Goldstein, Ozdenoren, and Yuan 2013). Thus, we expect to find larger effects where short-selling constraints are more likely to be binding.

Indeed, we find that the main effect is coming from small firms, and it is stronger for growth firms, firms with high discretionary accruals, firms with high dispersion in analysts' recommendations, and firms experiencing the largest negative market reaction or the highest trading volume around the disclosure of the pilot list. Firms in the pilot group with a big increase in short-selling activity exhibit larger reductions in investment and equity issuance. Moreover, we find that small firms in the pilot program decreased their equity issues significantly during the experiment. We also find that prices of pilot stocks are more sensitive to market-wide or firm-specific selling pressure. In general, all these results are consistent with predictions from both the overvaluation and

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bear raid hypotheses. Unfortunately, distinguishing between these theories is beyond the scope of our study.

Finally, we test whether managers learn about investment opportunities by looking at their stock prices, but the evidence indicates that managerial learning is not the main driver of our results. We also consider whether short sellers act as external monitors of corporate governance to mitigate the overinvestment problem. We find no evidence supporting this hypothesis either.

1. Sample, Data, and Variable Definitions

The SEC announced on July 28, 2004, a list of 968 firms to be included in the pilot group of Reg SHO, though the SEC selected the pilot firms at least a month earlier, on June 23, 2004, when the list was approved by the SEC board. The SEC selected firms from the Russell 3000 index listed on NYSE, NASDAQ, and AMEX and ranked them independently for each stock exchange by average daily traded volume. Every third firm on these lists was then included in the pilot group. Thus, the SEC experiment was a stratified random sample that ensured representation of the three markets and average trading volume. The objective of the pilot study was to test the impact of restrictions on short sales on the market volatility, price efficiency, and liquidity.[6]

We construct the main dataset from the Center for Research on Security Prices (CRSP). We build the Russell 3000 index based on market capitalization on May 28, 2004, and May 31, 2005. Consistent with the definition of the Russell 3000 at the reconstitution date, we exclude stocks with prices below \$1, pink sheet and bulletin board stocks, closed-end mutual funds, limited partnerships, royalty trusts, foreign stocks, and American Depositary Receipts (ADRs). In line with Diether, Lee, and Werner (2009) we keep firms that were in the Russell 3000 index in 2004 and 2005 and eliminate firms added to the index between June 2004 and June 2005, or firms that are deleted from the index due to acquisitions, mergers, or bankruptcies during the year. We merge this list with the list of pilot securities announced on July 28, 2004, by the SEC. Out of the 968 pilot securities in the initial list, 946 pilot securities remain in the sample after the first filter. Merging with Compustat and excluding utilities and financials leaves 1,930 firms (1,279 control, 651 pilot).[7] Our final sample is an unbalanced panel of 13,526 firm-year observations with 8,919 firm-year

[6] A first announcement was made on October 28, 2003 (Securities Exchange Act Release No. 48709), on the intention to carry out the experiment. External comments were requested. The final design of the experiment, the list of all firms in the pilot group, the group of firms for which all price tests were suspended, and the control group were announced on July 28, 2004 (Securities Exchange Act Release No. 50104).

The pilot program (Rule 202T) was part of a broader rule (Reg SHO), which was announced on the same day as Reg SHO and adopted on August 6, 2004 (Release No. 34–50103). It included provisions concerning location and delivery of short sales (Rule 203) aimed at reducing naked short selling, and new marking requirements for equity sales (Rules 200 and 201).

[7] Removing these filters does not qualitatively change our results.

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observations in the control group and 4,607 firm-year observations in the pilot group (an average of 1,690 firms per year, 576 of which are in the pilot group).

We use *Capital Expenditures*, *Changes in Total Assets*, and *Capital Expenditures plus R&D* as measures of corporate investment. *Capital Expenditures* is equal to investment in fixed assets (Compustat item *CAPX*) scaled by the beginning-of-the-year total assets (Compustat item *AT*). *Changes in Total Assets* is equal to the percent change in total assets. *Capital Expenditures plus R&D* is equal to investment in fixed assets (Compustat item *CAPX*) plus research and development expenses (Compustat item *XRD*) scaled by the beginning-of-the-year total assets.

We also construct *Equity Issues* and *Debt Issues* to measure the external financing activities of the firms during the Reg SHO experiment. *Equity Issues* is computed as the sale of common and preferred stock (Compustat item *SSTK*) scaled by beginning-of-the-year total assets. *Debt Issues* is computed as the long-term debt issues (Compustat item DLTIS) scaled by beginning-of-the-year total assets. Appendix 1 provides more detail on all of our variable definitions.

Table 1 presents summary statistics for all the firms in the sample.[8] Consistent with the random selection of the firms in the pilot and control group, we find no major differences between the two groups. For example, both groups of firms have roughly the same size, corporate spending, payout, and capital structure. None of the differences are statistically significant.[9] In Panel B, we find similar results for the subsample of small firms (below median assets). Thus, splitting the sample based on asset size (or any other measure of financial constraints) does not appear to induce a pre-treatment selection bias in our sample of pilot stocks.[10]

In auxiliary analysis, we also estimate probit and logit models that predict inclusion in the pilot group based on a comprehensive set of firm characteristics and find that none predict inclusion in the pilot. In short, our filtering process has not created any obvious sample selection bias, and the data support the hypothesis that our pilot group firms represent a random draw from the population.

2. Short Interest and Prices

In this section we reexamine the impact of Reg SHO on short interest levels and prices. Our focus in this section is primarily on prices because our results differ from previous studies. We follow the methodology in Diether, Lee, and

[8] To help mitigate the impact of outliers and measurement errors in the data, we winsorize or trim variables at the first and 99th percentiles.

[9] We also test for differences in the median for these variables using the two-sample Wilcoxon rank sum test, and find similar results.

[10] In Section 7.1 we show that our main results are robust to different measures of financial constraints beyond asset size.

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Table 1
Summary statistics

Panel A: Entire sample

	Pilot group				Control group					
	N	Mean	Median	Std. Dev.	*N*	Mean	Median	Std. Dev.	Diff.	*T*-stat.
Total Assets	651	3,448	735	11,276	1,279	3,845	693	12,786	−396	−0.67
Market-to-Book	651	2.31	1.77	1.68	1,270	2.24	1.70	1.65	0.07	0.90
Market Cap.	651	4,961	813	19,883	1,270	4,548	767	18,804	383	0.44
CAPX	643	5.82	3.78	6.60	1,249	5.59	3.45	6.40	0.23	0.74
Δ Total Assets	635	21.55	10.66	39.08	1,235	19.33	10.81	33.89	2.21	1.27
CAPXR&D	634	11.00	7.66	10.35	1,239	11.12	7.88	10.37	−0.12	−0.23
Cash Flow	641	9.15	10.22	14.81	1,250	8.59	10.23	16.86	0.56	0.72
Leverage	649	27.27	23.35	27.23	1,263	27.30	22.99	27.78	−0.03	−0.02
Equity Issues	625	5.62	1.42	14.17	1,202	5.52	1.40	13.87	0.10	0.15
Debt Issues	612	12.14	0.01	23.93	1,186	10.74	0.00	22.56	1.40	1.22
Dividends	642	0.90	0.00	1.80	1,248	0.87	0.00	1.78	0.03	0.35
Cash Holdings	643	26.87	16.35	31.49	1,250	27.47	15.62	31.00	−0.60	−0.40
Past Profitability	640	10.48	12.29	13.37	1,245	9.86	11.58	14.65	0.63	0.91

Panel B: Small firms (below-medianassets)

	Pilot group				Control group					
	N	Mean	Median	Std. Dev.	*N*	Mean	Median	Std. Dev.	Diff.	*T*-stat.
Total Assets	316	329	288	207	635	316	285	194	13	0.96
Market-to-Book	316	2.76	2.07	1.93	634	2.60	1.98	1.87	0.16	1.26
Market Cap.	316	588	393	510	634	542	354	472	46	1.56
CAPX	313	5.85	3.60	6.98	629	5.73	3.31	7.00	0.12	0.25
Δ Total Assets	306	29.25	13.73	49.27	622	25.78	13.93	41.67	3.47	1.12
CAPXR&D	310	13.66	10.27	12.25	626	14.02	10.47	12.26	−0.36	−0.42
Cash Flow	313	7.36	10.18	19.54	630	7.08	10.70	21.82	0.28	0.19
Leverage	316	18.62	5.01	27.34	630	17.64	4.96	25.33	0.98	0.55
Equity Issues	302	9.06	2.04	19.34	606	8.77	2.01	18.56	0.29	0.22
Debt Issues	296	11.52	0.00	24.49	603	9.99	0.00	23.86	1.52	0.89
Dividends	313	0.65	0.00	1.86	629	0.70	0.00	1.87	−0.05	−0.44
Cash Holdings	313	39.65	28.79	37.86	630	39.00	30.56	36.20	0.65	0.26
Past Profitability	312	7.63	10.88	16.94	629	6.92	10.36	18.57	0.70	0.56

Data are collected from the merged CRSP/Compustat Industrial database in the fiscal year that is the closest to July 28, 2004, the announcement date of the SHO pilot test. We exclude firms that are not in the Russell 3000 index in 2004 and 2005, financial services firms (SIC code 6000–6999), and regulated utilities (SIC code 4900). Small firms are the subset of firms that have total assets below the median. All variables are described in Appendix 1.

Werner (2009), but we focus on event windows around the announcement of the list of firms to be included in the Reg SHO experiment.

We focus on the announcement date because under rational expectations, investors should incorporate the future impact of the regulatory change at the time of the announcement. For example, Allen, Morris, and Postlewaite (1993) show that price bubbles can exist if investors are short-sale constrained either in the current period or in future periods. Even if all agents are rational and know the dividends with certainty, short-selling constraints can create a bubble by generating the belief among investors that they will be able to sell the stock to another investor at a higher price in the future. In this setting, an announcement of a future removal of short-selling constraints reduces current stock prices through recursive expectations as investors recognize that they will not be able to take advantage of other investors in the future. Therefore, this model predicts

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that the announcement of Reg SHO should have the same effect on prices as an immediate removal of the short-selling constraints.

In a slightly different setting, Scheinkman and Xiong (2003) explicitly calculate the value of this resale option and show that in the presence of short-selling constraints stock prices should incorporate the value of this option. Thus, the expected removal of short-selling constraints should be enough to lower current stock prices. Further, according to both models, investors should be more willing to short Reg SHO pilot stocks as long as the cost of short selling is smaller than the level of the expected overvaluation. Finally, Reg SHO could have an adverse effect on prices at the time of the announcement by increasing the incentives of bear raiders to manipulate the value of those firms that are more susceptible to short-selling activity (Goldstein and Guembel 2008; Goldstein, Ozdenoren, and Yuan 2013).

2.1 Short interest

In this section, we test whether the uptick rule is a binding constraint on short selling. In previous studies, Alexander and Peterson (1999) and Angel (1997) find that the uptick rule delays or prohibits the execution of most intraday short-sell orders. Chung (1991) also shows that the uptick rule constrains index arbitrage.

The pilot study for Regulation SHO was designed to answer whether or not the uptick rule and bid test are significant enough to really matter. Previous analyses document an increase in intraday short sales after the implementation of the pilot program on May 2, 2005 (e.g., Diether, Lee, and Werner 2009; Alexander and Peterson 2008, Office of Economic Analysis 2007). In this subsection we test whether *Short Interest*, a proxy for short-sales activity, increases after the announcement of the pilot program. As explained above, short sellers who anticipate a real effect of the suspension of price tests on pilot firms should increase their short-selling activity on these firms following the disclosure, even before the actual suspension of the price tests. To test this hypothesis, we construct a monthly time series of *Short Interest* from the monthly short interest reported by NASDAQ and NYSE. Monthly short interest is the number of all open short positions on the last business day on or before the 15th of each calendar month scaled by the previous month shares outstanding (from CRSP). We collect open short positions from NASDAQ and NYSE over the period 2002–2009.[11] To measure the unexpected component of short interest, we create the variable *Abnormal Monthly Short Interest*, which is equal to the residual of a firm fixed effect regression where the monthly short

[11] NASDAQ and NYSE brokers/dealers are required to report their short positions as of settlement on the 15th of each month or the preceding business day. It takes three business days to settle trades; therefore, the short interest number includes short sales that occurred three business days prior to the 15th.

Contrary to studies of Reg SHO that use daily short sales, we do not use these data because they are not available for the period preceding the implementation of the experiment. Instead, we use *Short Interest* and *Abnormal Short Interest* as proxy for short sales in our analysis.

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Table 2
SHO pilot and short interest

Panel A: All firms

	Short interest				**Abnormal short interest**			
	Before	After	Diff.	T-stat	Before	After	Diff.	T-stat
Pilot group	4.48	5.93	1.44[a]	(8.82)	0.04	0.20	0.16[c]	(1.76)
Control group	4.40	5.40	1.00[a]	(8.95)	−0.04	−0.12	−0.08	(−1.26)
Diff.	0.09	0.54[a]			0.08	0.32[a]		
T-statistic	(0.81)	(3.19)			(1.25)	(3.46)		
Diff.-in-diff.			0.45[b]	(2.26)			0.24[b]	(2.16)

Panel B: Small firms

	Short interest				**Abnormal short interest**			
	Before	After	Diff.	T-stat.	Before	After	Diff.	T-stat.
Pilot group	4.79	6.24	1.45[a]	(6.91)	−0.03	0.27	0.30[b]	(2.31)
Control group	4.73	5.67	0.94[a]	(6.43)	−0.03	−0.06	−0.03	(−0.30)
Diff.	0.07	0.58[a]			0.00	0.33[a]		
T-statistic	(0.43)	(2.80)			(−0.02)	(2.94)		
Diff.-in-diff.			0.51[b]	(1.99)			0.33[b]	(2.29)

This table presents mean values of *Short Interest* and *Abnormal Monthly Short Interest* for firms that were part of the pilot group and control group for one year before the announcement date (July 28, 2004) and for nine months after the announcement (but before the implementation date). *Short Interest* is the monthly mean ratio of net short positions outstanding reported on the 15th of each month to shares outstanding at the start of the month. *Abnormal Monthly Short Interest* is the residual of a firm fixed effect regression where *Short Interest* is regressed with firm fixed effects, controlling for month dummies, market-to-book, logarithm of lagged total assets, lagged return on assets, trading volume, and a dummy variable for listing on the NYSE. Averages are computed for all firms that are in the pilot group and in the control group, and for the subset of firms that have total assets below the median (small firms). T-statistics are constructed with Newey-West standard errors. [c,b,a] indicate significance levels of less than 10%, 5%, and 1%, respectively.

interest is regressed on month dummies, lagged market-to-book, logarithm of lagged total assets, lagged return on assets, trading volume, and a dummy variable for listing on the NYSE.

Table 2 presents both the average *Short Interest* and *Abnormal Monthly Short Interest* for the year before the announcement of the pilot test on July 28, 2004, and the nine months after the announcement. We stop the analysis on April 2005 to exclude any confounding effect from the actual implementation of Reg SHO. Panel A presents the results for all firms, and Panel B presents the results for small firms (below median total assets). We find that both *Short Interest* and *Abnormal Monthly Short Interest* increase more for firms in the pilot group. For example, when we examine the effect of Reg SHO on *Short Interest*, we find that the difference-in-differences are statistically significant and represent a relative increase of about 10% and 11% of the average monthly short interest for all firms and small firms, respectively. To put these numbers in perspective, Diether, Lee, and Werner (2009) report a magnitude of 8% around the implementation date using intraday data we do not have access to during the announcement period. The SEC's Office of Economic Analysis (OEA, 2007) finds that short-selling activity increased by two percentage points of total volume around the implementation date (or an 8% increase relative to the mean short-selling volume before the implementation date of 24%). Therefore,

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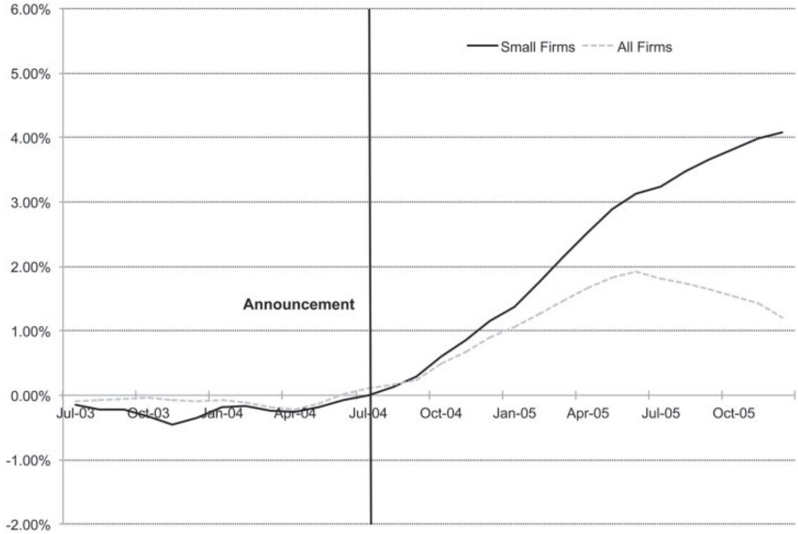

Figure 1
Cumulative abnormal short interest
This graph presents the cumulative monthly abnormal short interest for all the firms and small firms (total assets below the median) in the pilot group minus the average monthly abnormal short interest for the firms in the control group around the announcement date (July 28, 2004) of the pilot program. The first observable *Abnormal Monthly Short Interest* after the announcement date is for short position not settled on August 13, 2004. *Abnormal Monthly Short Interest* is the residual of a firm fixed effect regression where the monthly mean ratio of net short positions outstanding reported on the 15th of each month to shares outstanding at the start of the month is regressed on month dummies, market-to-book, logarithm of lagged total assets, lagged return on assets, trading volume, and a dummy variable for listing on the NYSE.

the effects add up to 19% taking into account both the announcement date and the implementation date.

To represent the trend in short-selling activity graphically, we plot the cumulative *Abnormal Monthly Short Interest* around the announcement date in Figure 1. This figure reveals a clear systematic increase in short-selling activity after the announcement of Reg SHO. Overall, our findings are consistent with the notion that the expected future removal of short-sales constraints creates incentives for investors to short-sell stocks in the current period.

2.2 The suspension of price tests, firm valuation, and stock returns

In this subsection, we test whether prices react to the announcement of the change in short-selling restrictions. As in many event studies, we examine a number of event windows. For example, the stock price reaction could occur (i) immediately after the approval (June 23, 2004), (ii) after the formal public announcement of the pilot experiment (July 28, 2004), or (iii) after the implementation of the experiment when the uptick rule is actually removed (May 2, 2005). We focus on the approval and announcement dates since these events should incorporate any change in expectations that might precede

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Table 3
Announcement day average daily abnormal returns

Panel A: All firms

Event window	[−1,1]		[0, 0]		[−10,1]		[−24,11]	
	Abnormal returns	T-stat.	Abnormal returns	T-stat.	Abnormal returns	T-stat.	Abnormal returns	T-stat.
Pilot	0.44	(0.81)	−0.65[a]	(−5.78)	−0.16	(−0.66)	−0.18	(−1.61)
Control	0.49	(0.84)	−0.68[a]	(−7.53)	−0.03	(−0.36)	−0.14	(−1.31)
Diff.	−0.06	(−1.47)	0.04	(0.27)	−0.13[a]	(−3.21)	−0.04[c]	(−1.67)
BHAR equivalent	−0.17		0.04		−1.52		−1.48	

Panel B: Small firms

Event window	[−1,1]		[0, 0]		[−10,1]		[−24,11]	
	Abnormal returns	T-stat.	Abnormal returns	T-stat.	Abnormal returns	T-stat.	Abnormal returns	T-stat.
Pilot	0.49	(0.63)	−1.02[a]	(−5.38)	−0.28	(−0.81)	−0.28[c]	(−1.78)
Control	0.80	(0.99)	−0.81[a]	(−5.59)	−0.08	(−0.24)	−0.19	(−1.27)
Diff.	−0.31[b]	(−2.46)	−0.21	(−0.88)	−0.20[a]	(−2.70)	−0.08[b]	(−2.03)
BHAR equivalent	−0.92		−0.21		−2.35		−3.05	

Panels A and B present the difference in equally weighted average daily abnormal returns of all firms in the sample and of the small firms in the sample (below-median total assets) around the announcement date of the list of pilot firms (July 28, 2004). We use four different time windows around the announcement date: [−1,1] (from July 27 to July 29, 2004), [0, 0] (July 28, 2004), [−10,1] (from July 14 to July 29, 2004), and [−24,11] (from June 23 to August 12, 2004). The daily abnormal returns are the difference between the firms' daily returns and the market's value weighted daily returns from CRSP. The BHAR equivalent is the buy-and-hold abnormal return compounded daily over the period considered. T-statistics are displayed within parentheses next to the relevant coefficient. Standard errors are clustered both at the firm and the date levels. [c,b,a] indicate significance levels of less than 10%, 5%, and 1%, respectively.

implementation. Further, we focus on this sample period because there is evidence of abnormal short-selling activity around the announcement date. Nevertheless, we test a wide variety of definitions and event windows to present a comprehensive review of price reactions.

First, we construct event study excess returns with various windows around the announcement and approval dates. Next, we also construct long-run abnormal returns for the sample period around the experiment (up to two years).

2.2.1 Event study CARs. Our first tests present simple event study cumulative abnormal returns around July 28, 2004, the date when the SEC announced the list of firms included in the pilot, and June 23, 2004, the date the SEC approved the list. We compute excess returns as the difference between the daily returns and the CRSP value-weighted returns. We then regress these returns on a dummy variable for the inclusion in the pilot, adjusting standard errors for heteroscedasticity and clustering them at both the firm and the date levels. This gives us the average daily excess returns around the announcement date and the difference in average daily excess returns between the pilot and the control firms.

Table 3 presents our analysis for four different event windows around the announcement date. In the first two columns, we report equally weighted buy

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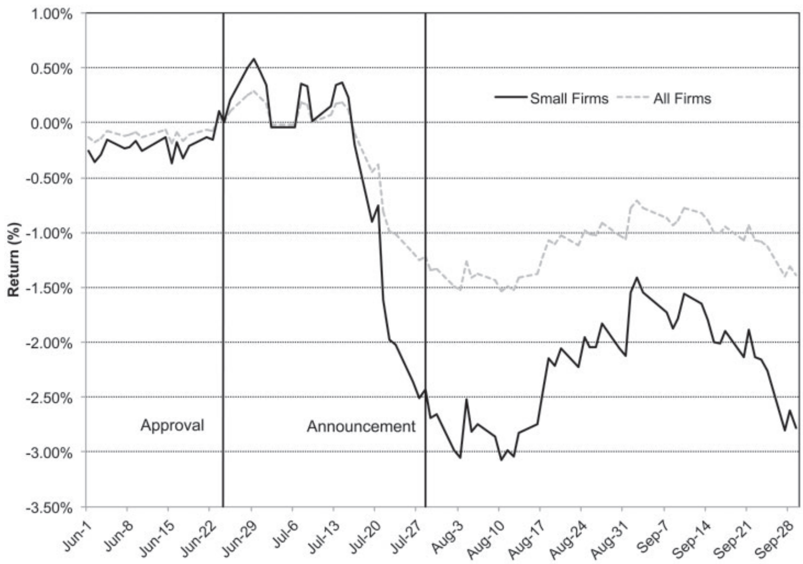

Figure 2
Buy-and-hold abnormal returns around the event dates
This graph presents the differences in equally weighted buy-and-hold abnormal returns between the pilot firms and the control firm for all firms in the sample and small firms around the approval date (June 23, 2004) and the announcement date (July 28, 2004) of the pilot program. Firms with stock prices below $5 on April 30, 2004, are excluded from the sample.

and hold abnormal returns (BHARs) for a three-day event window [−1, 1] and the event day itself [0, 0] around the official announcement day of July 28, 2004. Consistent with past studies (e.g., Diether, Lee, and Werner 2009), we find little movement in prices around the official announcement date.

The results change significantly when we expand the event window back toward the date that the SEC approved (but did not yet announce) the list of firms included in the pilot. Going back ten trading days before the public announcement [−10, 1], there is a significant decline in the prices of the pilot stocks. Over this time period, pilot stocks experienced a BHAR of −1.52% relative to the control group. Results are similar for a longer window around the announcement date [−24, 11]. Pilot firms experienced a BHAR of −1.48% relative to the control group. Moreover, we find that the price decline is larger for small firms. As shown in Panel B of Table 3, small firms experienced BHARs of −2.35% and −3.05% during the ten days before the announcement [−10, 1] and the longer window [−24,11], respectively.

A closer look at the difference in BHARs around the approval and announcement dates helps to explain the discrepancy between our results and past studies. Figure 2 plots the time series of the difference in BHARs between the pilot and control group around the announcement date. For both samples, it appears that the bulk of the negative abnormal performance occurred at

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Table 4
Long-horizon market-adjusted abnormal returns

Panel A: All firms

Event window	Six months		One year		Two years	
	Abnormal returns	T-stat.	Abnormal returns	T-stat.	Abnormal returns	T-stat.
Pilot	–1.43[c]	(–2.56)	–0.65	(–1.71)	–0.52[b]	(–2.48)
Control	0.00	(0.02)	–0.11	(–0.47)	–0.15[c]	(–1.15)
Diff.	–1.43	(–1.50)	–0.54	(–1.04)	–0.37	(–1.29)
BHAR equivalent	–8.89		–6.68		–9.27	

Panel B: Small firms

Event window	Six months		One year		Two years	
	Abnormal returns	$T-$stat.	Abnormal returns	T-stat.	Abnormal returns	T-stat.
Pilot	–1.74[c]	(–2.38)	–1.28[a]	(–3.16)	–0.84[a]	(–2.78)
Control	–0.92	(–1.18)	–0.57	(–1.20)	–0.49[c]	(–1.75)
Diff.	–0.82[a]	(–6.49)	–0.71[a]	(–4.08)	–0.35[b]	(–2.08)
BHAR equivalent	–5.02		–8.20		–8.75	

Panels A and B present the difference in abnormal value-weighted monthly alphas of portfolios, computed using the market model. The alphas are computed for all firms in the pilot group and in the control group (Panel A) and firms in the pilot group and in the control group that have total assets below the median value of total assets (small firms in Panel B). We compute alphas over six-month, one-year, and two-year periods after June 23, 2004. The BHAR equivalent is the buy-and-hold abnormal return compounded monthly over the period considered. Observations with lagged stock prices below \$5 are excluded from the sample. T-statistics are displayed within brackets next to the relevant coefficient. Standard errors are adjusted for heteroscedasticity. [c,b,a] indicate significance levels of less than 10%, 5%, and 1%, respectively.

least two weeks before the announcement date. The constrained firm sample exhibits a negative reaction starting on July 14, 2004, but the relative negative performance persists more or less one week after the official announcement date up to -3% around the first two weeks of August 2004.[12] Results are similar using value-weighted portfolios.

2.2.2 Long-run returns. We also test whether the negative price reaction to the announcement of the pilot firms in Reg SHO persists in the long run. We construct abnormal returns six months, one year, and two years after the announcement of the pilot program. We form separate portfolios for the pilot and control groups, and then compute the value-weighted return using monthly CRSP returns.[13] We then estimate regressions of excess returns against a market factor (CRSP value-weighted index returns) and collect the alphas to tests whether long-run returns underperform the market.

Table 4 reports our results. Although we do not find any statistically significant abnormal returns using all firms in the sample (Panel A), the economic magnitude of the point estimate is large at -9.27% buy-and-hold

12 We have been unable to identify patterns of insider trading that could explain this early stock market reaction for the pilot firms using CEOs' stock transactions reported to the SEC in form 144. We perform and discuss an analysis of trade level data in Section 7.3 using TAQ data.

13 Since there is evidence of price effects after the approval of Reg SHO on June 23, 2004, we adjust the monthly returns from CRSP on July 2004 to include the returns from June 23, 2004, to June 30, 2004.

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abnormal returns over two years. For our sample of small firms in the pilot group, the results are significant (Panel B). The difference in monthly abnormal returns ranges from −0.82% when the holding period is six months (−5% buy-and-hold abnormal returns) to −0.35% over two years (−8.75% buy-and-hold abnormal returns). Both the economic magnitude and the statistical significance of these results are large, suggesting that the impact of the regulatory change is stronger for our sample of small firms.[14]

3. The Real Effects of the Suspension of Price Tests

In the previous section, we show that Reg SHO results in more short-selling activity and lower prices for firms in the pilot group, especially small firms. Since the selection process is random, the negative shock to equity values is independent of the investment opportunities. In this section, we use the Reg SHO natural experiment to identify a causal link between the removal of short-selling constraints and corporate policies.

3.1 Univariate tests

Table 5 presents univariate tests of whether pilot firms reduce *Capital Expenditures*, *Total Assets*, and *Capital Expenditures plus R&D* relative to control firms around the experiment. There is no difference in these measures prior to the pilot experiment. However, firms in the pilot group invest less than the control group during the experiment. The effects are statistically significant and stronger for small firms. For example, the difference-in-differences in *Capital Expenditures* is −0.60 percentage points of total assets for all the firms in the sample and −0.97 percentage points for the small firms in the sample. The latter result corresponds to a relative reduction of about 17% relative to the mean *Capital Expenditures* for the small firms in the sample. This represents about 14% of the yearly standard deviation of total investment for the small firms in the sample.

The difference-in-differences in *Changes in Total Assets* corresponds approximately to a reduction of 21% relative to the mean for the small firms in the sample. Although this reduction seems economically large, it is only 12% of a yearly standard deviation of *Changes in Total Assets*. The difference-in-differences of *Capital Expenditures plus R&D expenses* is also negative and statistically significant for the small firms.

Our univariate evidence is mixed with respect to the financing variables. The difference-in-differences are consistently negative for both *Equity Issues* and *Debt Issues,* but the statistical evidence is weak. In the full sample, the decline in *Debt Issues* is marginally significant, while the decline in *Equity Issues* is marginally significant only in the small sample. Nevertheless, it is interesting

[14] We do not find any evidence that the long-term returns are driven by outliers. In fact, our results become slightly stronger when we truncate the distribution of monthly stock returns at the first and 99th percentiles.

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Table 5
Real effects of the SHO pilot program: Univariate results

Panel A: All firms

	CAPX		ΔTotal Assets		CAPXR&D		Equity Issues		Debt Issues	
	Before	After	Before	After	Before	After	Before	After	Before	After
Pilot	5.63	5.82	14.90	12.68	10.85	10.77	4.68	3.45	11.00	10.54
Control	5.37	6.16	13.20	14.14	10.93	11.40	4.70	3.99	9.79	10.71
Diff.	0.26	−0.34	1.71	−1.45	−0.09	−0.63	−0.03	−0.54	1.21	0.17
(pilot - control)	(0.94)	(−1.07)	(1.19)	(−1.50)	(−0.17)	(−1.35)	(−0.06)	(−1.46)	(1.49)	(−0.67)
Diff.-in-diff.		−0.60[a]		−3.16[b]		−0.54		−0.52		−1.38[c]
		(−2.85)		(−2.28)		(−1.62)		(−1.10)		(−1.67)

Panel B: Small firms

	CAPX		ΔTotal Assets		CAPXR&D		Equity Issues		Debt Issues	
	Before	After	Before	After	Before	After	Before	After	Before	After
Pilot	5.46	5.40	18.25	14.07	13.49	12.56	7.59	4.69	8.69	8.31
Control	5.31	6.21	15.31	17.25	13.77	13.88	7.21	5.92	8.34	9.74
Diff.	0.15	−0.82[c]	2.93	−3.18[b]	−0.28	−1.33[c]	0.38	−1.23[c]	0.35	−1.46
(pilot -control)	(0.37)	(−1.71)	(1.50)	(−2.04)	(−0.55)	(−1.76)	(0.43)	(−1.84)	(0.31)	(−1.22)
Diff.-in-diff.		−0.97[a]		−6.12[b]		−1.05[c]		−1.61[c]		−1.80
		(−2.88)		(−2.55)		(−1.81)		(−1.82)		(−1.52)

This table presents the mean values of *Capital Expenditures, Changes in Total Assets, CAPXR&D, Equity Issues,* and *Debt Issues* three years before and three years after the announcement of the pilot program for all firms in the sample (Panel A) and for small firms (total assets below the median – Panel B). The *Diff.-in-diff.* measures the change in the mean value after the announcement and during the pilot (versus before the announcement) for firms in the pilot group relative to firms in the control group. Point estimates are based on an OLS regression where investment is regressed on a dummy for firms in the pilot, a dummy variable equal to one after the pilot started and the interaction term of these two variables. *Before* is the period that extends three years before the announcement of the pilot program (July 28, 2004), while *After* is the three-year period of the pilot program before the repeal of the uptick rule (July 28, 2004, to July 6, 2007). We require the fiscal year to overlap at least six months during the period considered to be included in the sample. T-statistics are displayed within parentheses next to the relevant coefficient. Standard errors are adjusted for heteroscedasticity and clustered at the firm level. All variables are detailed in Appendix 1. [c,b,a] indicate significance levels of less than 10%, 5%, and 1%, respectively.

to note the consistent negative point estimates, which suggest that firms in the treatment may have reduced security issuance relative to the control group.

3.2 Multivariate tests

While the results in the previous subsection provide an overview of mean differences in corporate behavior between our treatment and control samples, there are some reasons to be cautious about these results. Estimates for the pre- and post-Reg SHO period are formed over three years of annual data, which may induce some confounding effects if the randomness of the pilot group diminishes over time as corporate actions change.

In this section, we account for these potential concerns by testing whether inclusion in the pilot program had an effect on corporate actions after accounting for firm fixed effects, year fixed effects, and variations in firm size, cash flows, profitability, and other characteristics. In this regression setting, our identification strategy relies on the exogenous shock of removing short-selling constraints on real decisions. We measure the effects of Reg SHO with a *Treatment* dummy variable equal to one if the firm was announced to be in the pilot group or has had price tests suspended on its stock for at least six

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months during the fiscal year end-date, and equal to zero otherwise.[15] The SEC announced the list of firms in the SHO pilot group on July 28, 2004, and suspended price tests on May 2, 2005, for firms in the pilot group, while it suspended price tests for all firms on July 7, 2007.

Panel A of Table 6 presents our regression results. The first set of regressions shows that *Capital Expenditures* decline after the introduction of Reg SHO and the effect is concentrated in small firms. The economic magnitude of the effect is again large: the effect of Reg SHO on *Capital Expenditures* is a reduction between 10% and 11% of mean *Capital Expenditures* (15% and 18% of the yearly within firm standard deviation of *Capital Expenditures*). There is also evidence that Reg SHO adversely affected *Changes in Total Assets* and *Capital Expenditures plus R&D Expenses.* These effects are primarily concentrated in small firms.

Given the recent evidence indicating that financial constraints affect corporate investment and financing decisions (e.g., Campello, Graham, and Harvey 2010; Campello, Giambona, Graham, and Harvey 2011), we expect small firms to be the most affected by Reg SHO. We test this assumption using *Equity Issues* and *Debt Issues* as the dependent variables. We control for the firms' cash flows, size, profitability, and lagged leverage. The results are presented in Panel B of Table 6.

Small firms that are subject to the suspension of price tests decrease equity issuance activity but do not decrease debt issuance. The coefficient on the *Treatment* dummy variable or the interacted *Small Firms* and *Treatment* variable is negative and significant at the 5% or 1% level in all specifications. The economic magnitude of the reduction is large: from 24% to 40% of mean *Equity Issues* but only 11% to 19% of the yearly within firm standard deviation. The point estimate on *Debt Issues* is negative but statistically insignificant for small firms, but significant for all firms. We conclude that Reg SHO caused a reduction in financing activity, especially in the equity issuance activity of small firms.

Our results are consistent with an increased cost of equity issuance due to a negative shock to stock prices (Gilchrist, Himmelberg, and Huberman 2005). Further, as the pilot stocks become more sensitive to both market-wide and firm-specific negative news (see Section 5), rational managers may become more reluctant to engage in corporate decisions (e.g., issue equity) that, on average, elicit negative stock returns (e.g., Asquith and Mullins 1986; Bayless and Chaplinsky 1996; Ritter 2003) and/or create incentives for bear raids (Goldstein, Ozdenoren, and Yuan 2013).

We also note that although the magnitudes of the effects in our paper appear to be large, they are generally in line with magnitudes documented in other

15 Given that we include firm fixed effects, this specification is equivalent to interacting a dummy for the inclusion in the pilot group with a dummy for the time period over which the experiment is conducted. The only difference is that we allow firms in the control group that are later subject to the repeal of the uptick rule from July 7, 2007, to be considered as part of the experiment.

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Table 6
Corporate investment, financing, and the SHO pilot program: Multivariate results

Panel A: Corporate investment

	CAPX		Δ Total Assets		CAPX R&D	
	1	2	3	4	5	6
Treatment	−0.54[b]	−0.01	−4.10[b]	0.02	−0.85[c]	−0.10
	(−2.00)	(−0.06)	(−2.13)	(0.01)	(−1.91)	(−0.40)
Treatment × Small Firms		−0.62[a]		−2.34[c]		−0.76[b]
		(−2.63)		(−1.93)		(−2.05)
Cash Flow	0.03[a]	0.03[a]	0.58[a]	0.57[a]	−0.07[a]	−0.07[a]
	(3.48)	(4.83)	(6.91)	(8.24)	(−3.80)	(−4.38)
Log(Lagged Total Assets)	−0.88[b]	−0.96[a]	−36.15[a]	−34.68[a]	−5.82[a]	−5.06[a]
	(−2.35)	(−3.74)	(−14.70)	(−19.91)	(−7.49)	(−9.86)
Past Profitability	3.68[a]	5.23[a]	20.32[c]	28.20[a]	0.07	3.03
	(3.80)	(6.39)	(2.33)	(3.83)	(0.04)	(1.92)
Sample	Small	All	Small	All	Small	All
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
N	5,807	11,929	5,812	11,942	5,807	11,929
Adj. R^2	0.70	0.74	0.31	0.30	0.78	0.79

Panel B: Financing

	Equity Issues		Debt Issues	
	1	2	3	4
Treatment	−1.43[b]	0.73[b]	−1.09	−1.98[b]
	(−1.97)	(1.96)	(−0.91)	(−2.21)
Treatment × Small Firms		−2.45[a]		1.42
		(−4.62)		(1.27)
Cash Flow	−0.07[c]	−0.04	−0.02	−0.03
	(−1.79)	(−1.57)	(−0.53)	(−0.95)
Log(Lagged Total Assets)	−10.34[a]	−7.66[a]	−3.93[a]	−5.86[a]
	(−9.71)	(−11.07)	(−3.46)	(−6.08)
Past Profitability	2.92	−0.23	4.09	9.13[b]
	(0.68)	(−0.06)	(0.90)	(2.33)
Lagged Leverage	0.08[a]	0.05[a]	−0.07[b]	−0.09[a]
	(4.09)	(3.65)	(−2.48)	(−4.47)
Sample	Small	All	Small	All
Firm fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
N	5,560	11,340	5,363	11,099
Adj. R^2	0.42	0.42	0.44	0.48

This table presents the results of OLS regressions with firm fixed effects and year fixed effects for the subset of small firms (total assets below the median) and all firms in the sample. Panel A reports the results for *CAPX, Changes in Total Assets,* and *CAPXR&D,* while Panel B reports the results for *Equity Issues* and *Debt Issues*. *Treatment* is a dummy variable equal to one if the company is in the pilot group of Reg SHO and the fiscal year includes at least six months of activity after the announcement of Reg SHO (July 28, 2004) and equal to zero otherwise. The dummy variable is equal to zero for firms in the control group before the suspension of all prices for all firms in the U.S. stock markets and equal to one after the firm's fiscal year includes at least six months after the suspension of all prices for all firms in the U.S. stock markets (July 6, 2007). T-statistics are displayed within parentheses under each coefficient. Standard errors adjust for heteroscedasticity and within correlation clustered by firm. All variables are defined in Appendix 1. [c,b,a] indicate significance levels of less than 10%, 5%, and 1%, respectively.

studies on the real effects of share price changes. The short-term announcement BHARs are similar in magnitude to the ones reported in previous studies on important corporate announcements, such as dividend changes and equity

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issues. Further, our results are consistent with the broad range of investment responses documented in previous studies. For example, Chirinko and Schaller (2001) find a one-sigma change in their measure of misvaluation is associated with a 20–60% change in investment relative to the mean. Goyal and Yamada (2004) find similar declines in investment after the correction of an asset price bubble in Japan. Campello, Ribas, and Wang (2014) find a 50% change in investment after a shock to stock market activity in China. More recently, Derrien and Kecskés (2013) find that after a decline in analyst coverage—an event that elicits an abnormal market reaction of roughly –1%—firms reduce investment by 14% relative to median investment. To put these numbers in perspective, we find a 10% change in investment relative to mean investment after permanent price changes of roughly 3% in the short run and 10% in the long run. Overall, our results do not appear to be outliers with respect to the magnitudes reported in other studies.

3.3 Linking real effects to asset prices and short-selling activity

In order to link changes in investment policy directly to Regulation SHO price and short-selling effects, we test whether variation in corporate behavior can be explained by variations in short-run event study returns, long-run abnormal returns, trading volume, and short-selling activity. We expect firms that exhibited the most negative stock returns and the highest trading volume and short-selling activity around the announcement date to be the most impacted by the experiment.

To test this hypothesis, we replicate the analysis in Table 6 interacting the Reg SHO dummy with the following variables: *Low CAR*, *Low BHAR*, *High Trading Volume*, and *High Short Interest*. *Low CAR* is a dummy variable equal to one if the announcement daily abnormal returns of the firm 12 trading days around the announcement date of the pilot program (July 14, 2004, to July 29, 2004) fall in the bottom quintile and equal to zero otherwise. *Low BHAR* is a dummy variable equal to one if the two-year buy-and-hold abnormal returns after June 23, 2004, falls in the bottom quintile and equal to zero otherwise. *High Trading Volume* is a dummy variable equal to one if the sum of the common stock daily traded volume (from CRSP) divided by 100 over the twelve trading days around the announcement date of the pilot program (July 14, 2004, to July 29, 2004) falls in the top quintile and equal to zero otherwise. *High Short Interest* is a dummy variable equal to one if the average reported monthly short interest scaled by the total number of shares outstanding during the fiscal year falls in the top quintile and equal to zero otherwise.

Table 7 reports the results from this analysis using the sample of small firms. We find that firms with the lowest event CARs around the announcement date experience the largest decreases in investment and security issuance. For example, pilot firms in the lowest quintile of event study CARs experience a significant reduction in *Total Assets*, *Capital Expenditures plus R&D Expenses*, and *Equity Issues* relative to the firms in the other quintiles. This table also shows

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Table 7
Linking the real decisions to asset prices and short-selling activity: Multivariate results for small firms

	Low CAR		Low BHAR		High Trading Volume		High Short Interest	
	Treatment Dummy	Interaction	Treatment Dummy	Interaction	Treatment Dummy	Interaction	Treatment Dummy	Interaction
CAPX	−0.43[c]	−0.37	−0.05	−1.66[a]	−0.06	−0.32[a]	−0.09	−1.57[a]
	(−1.86)	(−1.18)	(−0.20)	(−5.15)	(−0.26)	(−6.43)	(−0.38)	(−4.90)
Δ Total Assets	−1.76	−7.68[a]	1.55	−20.46[a]	−1.78	−1.30[a]	−2.58	−5.28[b]
	(−0.94)	(−3.64)	(0.82)	(−7.65)	(−0.98)	(−3.23)	(−1.38)	(−2.02)
CAPXR&D	−0.61[c]	−0.77[c]	0.74	−4.46[a]	−0.16	−0.38[a]	−0.23	−1.73[a]
	(−1.77)	(−1.65)	(0.84)	(−8.62)	(−0.48)	(−5.05)	(−0.67)	(−3.57)
Equity Issues	−0.64	−2.59[b]	0.50	−6.57[a]	−0.70	−0.48[a]	−0.54	−2.62[b]
	(−0.77)	(−2.30)	(0.72)	(−5.30)	(−0.87)	(−2.73)	(−0.67)	(−2.35)
Debt Issues	−0.22	−2.77[c]	−0.53	−2.47	−0.89	−0.12	0.10	−3.84[b]
	(−0.20)	(−1.84)	(−0.47)	(−1.54)	(−0.82)	(−0.52)	(0.09)	(−2.49)

This table presents the results of OLS regressions with firm fixed effects and year fixed effects for the small firms in the sample (firms with total assets below the median). The dependent variables are *CAPX, Changes in Total Assets, CAPXR&D, Equity Issues,* and *Debt Issues*. Controls are identical to those used in the main specification of the multivariate analysis (Table 6, Panels A and B). We interact the *Treatment* dummy (*Reg SHO* dummy) with the following variables: *Low CAR, Low BHAR, High Trading Volume*, and *High Short Interest. Low CAR* is a dummy variable equal to one if the announcement daily abnormal returns of the firm twelve trading days around the announcement date of the pilot program (July 14, 2004, to July 29, 2004) falls in the bottom quintile and equal to zero otherwise. *High BHAR* is a dummy variable equal to one if the two-year buy-and-hold abnormal returns after June 23, 2004, falls in the bottom quintile and equal to zero otherwise. *High Trading Volume* is a dummy variable equal to one if the sum of the common stock daily traded volume (from CRSP) over the twelve trading days around the announcement date of the pilot program (July 14, 2004, to July 29, 2004) falls in the top quintile and equal to zero otherwise. *High Short Interest* is a dummy variable equal to one if the lagged average reported monthly short interest scaled by the total number of shares outstanding during the fiscal year falls in the top quintile, and is equal to zero otherwise. We report only the coefficient estimates of the *Treatment* dummy variable and the interaction term for brevity. T-statistics are displayed within parentheses under each coefficient. All variables are defined in Appendix 1. [c, b, a] indicate significance levels of less than 10%, 5%, and 1%, respectively.

that long-run returns for pilot stocks are correlated with changes in corporate behavior.

Our results also show up in an analysis of trading activity around the announcement date. Table 7 shows that firms experiencing the greatest increase in trading volume around the announcement window also experience the largest decrease in future investment and equity issues, while the firms that exhibit increased short-selling activity after the announcement and during the experiment decrease more their investment, equity issues, and debt issues.

Overall, we find a strong association between stock price changes and corporate decisions for our random sample of pilot stocks. Even within our random sample of pilot stocks, we find that stocks with a greater effect on prices and trading activity experience a greater reaction in real corporate activity. These results are consistent with both the overvaluation and bear-raids hypotheses.

4. Cross-sectional analysis

In this section, we perform a series of tests that identify the source of the impact on corporate behavior. We find that the real effects of Reg SHO are concentrated

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among the firms most likely to be impacted by Reg SHO based on a priori characteristics. On the one hand, the overvaluation hypothesis predicts that the effect of Reg SHO should be stronger on pilot firms that were more likely to be overvalued. On the other hand, the bear-raids hypothesis suggests that the effect should be stronger on pilot firms that are more likely to be susceptible to the real effects of bear raiders. Following the intuition in Goldstein and Guembel (2008), one would expect bear raiders to target firms that are more likely to cancel profitable investment projects (e.g., high-growth firms) and firms with high levels of asymmetric information.

We test these hypotheses using *High Discretionary Accruals*, *High Market-to-Book*, and *High Analyst Dispersion*. *High Discretionary Accruals* is a dummy variable equal to one if the *Discretionary Accruals* for the firm fell into the top quintile before the announcement of Reg SHO. *Discretionary Accruals* represent the abnormal portion of total accruals. While Polk and Sapienza (2009) and Hirshleifer et al. (2011) argue that accruals are a proxy for overvaluation as managers use accruals to inflate the non-cash component of their earnings, Wu, Zhang, and Zhang (2010) argue that high discretionary accruals reflect high-growth opportunities.

High-Market-to-Book is a common proxy for both overvaluation and investment opportunities (e.g., Tobin 1969; Baker, Stein, and Wurgler 2003). High-growth firms may be more likely to be affected by bear raids because their investment is more sensitive to prices. We use a dummy variable equal to one if the firm's market-to-book ratio falls into the top quintile before the Reg SHO announcement date and equal to zero otherwise.

We use stock analysts' disagreement as a proxy for investors' dispersion of beliefs. While firms with large investors' disagreement are more likely to be overvalued in the presence of short-selling constraints (Diether, Malloy, and Scherbina 2002), the dispersion may also reflect more uncertainty about investment opportunities, which theory predicts should increase the probability of bear raids (Goldstein and Guembel 2008). We use a dummy variable equal to one if the analysts' standard deviation of recommendations falls in the top quintile before the Reg SHO announcement date and equal to zero otherwise.

The results from this analysis are reported in Table 8. Consistent with our predictions, firms that are likely to be overvalued and/or more susceptible to bear raids before the announcement date of Reg SHO decrease their *Capital Expenditures* and *Capital Expenditures plus R&D Expenses* based on both discretionary accruals and market-to-book ratio. They also reduce their *Total Assets* based on market-to-book. Firms with high analyst dispersion show a bigger reduction in *Changes in Total Assets* and *Capital Expenditures plus R&D Expenses.* Firms in the treatment group with high market-to-book or analyst dispersion reduce *Equity Issues* more than the control group. We do not find any effects on *Debt Issues*. Overall, our evidence suggests that the effect of Reg SHO on real decisions is stronger in subsamples often associated with mispricing and/or susceptibility to bear raids.

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Table 8
Cross-sectional analysis for small firms

	High Discretionary Accruals		High Market-to-Book		High Analyst Disp.	
	Treatment Dummy	Interaction	Treatment Dummy	Interaction	Treatment Dummy	Interaction
CAPX	−0.41[c]	−0.74[b]	0.01	−1.73[a]	−0.49[b]	−0.37
	(−1.85)	(−2.01)	(0.04)	(−5.64)	(−2.25)	(−0.90)
ΔTotal Assets	−4.03[b]	1.51	−2.24	−4.72[c]	−2.42	−9.66[a]
	(−2.26)	(0.51)	(−1.19)	(−1.89)	(−1.37)	(−2.89)
CAPXR&D	−0.51	−1.28[b]	0.10	−2.63[a]	−0.50	−1.76[a]
	(−1.52)	(−2.32)	(0.29)	(−5.66)	(−1.50)	(−2.85)
Equity Issues	−1.50[c]	0.41	−0.70	−2.27[b]	−1.00	−3.17[b]
	(−1.89)	(0.31)	(−0.84)	(−2.06)	(−1.27)	(−2.15)
Debt Issues	−1.01	−0.39	−1.41	1.01	−1.03	−0.45
	(−0.94)	(−0.22)	(−1.24)	(0.68)	(−0.97)	(−0.23)

This table presents the results of the same OLS regressions as those presented in Table 6 for small firms in the sample (firms with total assets below the median). The dependent variables are *CAPX, Changes in Total Assets, CAPXR&D, Equity Issues,* and *Debt Issues. High Discretionary Accruals* is a dummy variable equal to one if *Discretionary Accruals* before the announcement of the SHO pilot program falls in the top quintile, and is equal to zero otherwise. *High Market-to-Book* is a dummy variable equal to one if the *Market-to-Book* before the announcement of the SHO pilot program falls in the top quintile, and is equal to zero otherwise. *High Analyst Dispersion* is a dummy variable equal to one if the standard deviation of analyst recommendations before the announcement of the SHO pilot program falls in the top quintile and is equal to zero otherwise. We report only the coefficient estimates of the *Treatment* dummy variable and the interaction term for brevity. All variables are defined in Appendix 1. [c,b,a] indicate significance levels of less than 10%, 5%, and 1%, respectively.

5. Downside risk and sensitivity to news announcements

We also look at the daily returns during bearish stock market days and around negative earnings announcements to examine whether the firms in the pilot group became more sensitive to negative information. According to the overvaluation hypothesis, if fewer pessimistic investors are sidelined after the removal of the constraints, then firms in the pilot should react more to bad news (Chen, Hong, and Stein 2002; Hong and Stein 2003). Alternatively, bear raids may be more likely during times of heavy selling pressure, which may amplify the sensitivity of prices to bad news. Because small firms are more sensitive to the change in short-sales regulation, we expect stronger results for these firms.

Our first set of tests examines the effect of bad market days on stock returns. To perform this analysis, we first sort daily market returns into quintiles and test whether the raw returns of firms in the pilot group are more negative in bad market days (quintile 1) after the announcement of the pilot program than before relative to the control group. This difference-in-differences test reveals whether pilot firms become more sensitive to bad news than control firms after the announcement of Reg SHO.

Table 9 presents the results from our difference-in-differences tests. The two groups of firms do not have different returns on bad market days before the announcement of Reg SHO. However, firms in the pilot group display more negative returns than the control firms after the announcement in the

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Table 9
Sensitivity to daily market returns

Panel A: All firms

	Before				After					
Quintile	Pilot	Control	Diff.	T-stat.	Pilot	Control	Diff.	T-stat.	Diff.-in-Diff.	T-stat.
1 (Low)	−1.44	−1.48	0.04	(1.06)	−1.41	−1.34	−0.07[b]	(−2.37)	−0.11[a]	(−3.76)
2	−0.26	−0.25	−0.00	(−0.22)	−0.37	−0.33	−0.04[b]	(−2.49)	−0.04	(−1.56)
3	0.23	0.23	0.00	(0.08)	0.10	0.12	−0.01	(−1.25)	−0.02	(−0.85)
4	0.75	0.78	−0.03	(−1.48)	0.59	0.58	0.01	(0.50)	0.04	(1.61)
5 (High)	1.62	1.60	0.01	(0.40)	1.34	1.31	0.02	(1.05)	0.01	(0.37)

Panel B: Small firms

	Before				After					
Quintile	Pilot	Control	Diff.	T-stat.	Pilot	Control	Diff.	T-stat.	Diff.-in-Diff.	T-stat.
1 (Low)	−1.56	−1.64	0.08	(1.61)	−1.58	−1.50	−0.09[b]	(−2.14)	−0.17[a]	(−3.88)
2	−0.24	−0.25	0.01	(0.32)	−0.41	−0.35	−0.05[b]	(−1.96)	−0.06	(−1.60)
3	0.25	0.26	−0.02	(−0.67)	0.08	0.10	−0.02	(−1.19)	−0.01	(−0.24)
4	0.86	0.90	−0.04	(−1.30)	0.62	0.62	−0.00	(−0.18)	0.03	(0.78)
5 (High)	1.79	1.77	0.02	(0.33)	1.44	1.45	−0.01	(−0.20)	0.03	(0.51)

Panels A and B present the mean daily raw returns for all firms in the sample (Panel A) and the small firms in the sample (below-median assets, Panel B) that were part of the pilot experiment, and firms that were part of the control group. We sort the observations by quintiles based on the value-weighted daily market returns (from CRSP), and then compute the average daily market returns for the pilot and control firms for each quintile. Quintile 1 of the value-weighted daily market returns is the lowest quintile of market daily returns while quintile 5 is the largest. The *Diff.-in-diff.* measures the change in mean daily returns after the announcement of the pilot (versus before the announcement of the pilot) for the pilot group relative to the control group. Point estimates are based on OLS regressions where the daily returns are regressed on a dummy for firms in the pilot, a dummy variable equal to one after the experiment was approved by the board of the SEC (June 23, 2004), and the interaction term of these two variables. *Before* is the one-year period before June 23, 2004. *After* is the one-year period after June 23, 2004. Standard errors are clustered both at the firm and date levels. [c, b, a] indicate significance levels of less than 10%, 5%, and 1%, respectively.

lowest quintile only: the worst market days. The difference-in-differences is statistically significant at the 1% level, and economically large: returns are 8% more negative in bad market days for pilot firms in the sample, and 11% more negative for small pilot firms.

We also test whether there are changes in the sensitivity to firm-specific news. We test for differences between pilot and control group reactions to earnings news using earnings surprises relative to I/B/E/S quarterly consensus analyst forecasts. The results are not reported in the interest of space. On average, small firms in the pilot group show larger negative abnormal returns relative to the control group on negative earnings news after the passage of Reg SHO. We do not find any corresponding reaction to positive earnings news. This is consistent with the hypothesis that increased short selling puts more downward pressure on stocks with bad news. All of our results point to a tangible downside price effect on small firms in the pilot group. This effect seems to be driven by increased short-selling activity even before the removal of short-selling constraints, and by large informed sell trades by shareholders of the small firms included in the pilot group.

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6. Alternative Explanations

One potential explanation for our results is that short sellers discipline managers by monitoring their actions through short-selling behavior. By shorting stocks of mismanaged firms, short sellers may put pressure on managers to alter their corporate financial policies in a way that is consistent with better corporate governance. Admati and Pfleiderer (2009) argue that the threat by large shareholders of selling shares may be credible and may discipline managers. If this hypothesis is true, then Reg SHO, which reduced the cost of short selling, could positively affect stock prices by incentivizing managers to be more efficient. However, the negative market reaction surrounding the announcement of Reg SHO that we document does not support this view.

We also test whether managerial learning can explain our results. Chen, Goldstein, and Jiang (2007) document that managers use information impounded into stock prices to improve corporate investment decisions. Therefore, an alternative explanation is that managers learn in an environment where prices go down on average, and therefore reduce investment. To test this hypothesis, we reestimate Chen, Goldstein, and Jiang's (2007) corporate investment specification using the Reg SHO experiment as an instrument for increased stock price informativeness. However, we do not find that firm's investment becomes more sensitive to stock prices after the experiment, which suggests that managerial learning may not be the primary channel in our sample.

Our results are robust to the exclusion of firm-year observations with a fiscal year end date after November 31, 2007. This suggests that our results are not driven by the consequences of the financial crisis. It is also possible that the uncertainty generated by Reg SHO increased firms' incentives to postpone their capital projects by increasing the value of the option to wait. However, we do not find any evidence that the reduction in investment and equity issues is due to an increase in firms' idiosyncratic volatility.

Another interpretation of our results is that Reg SHO increased the visibility of pilot stocks, leading to a Hawthorne effect.[16] That is, firms may change their behavior because they are being observed. Such a "placebo effect" may cause firms to change their behavior, regardless of how binding the uptick rule may have been. However, in univariate difference-in-differences tests, we find that the repeal of the uptick rule in July 2007 leads to a more pronounced reduction of investment among small control firms that are no longer subject to short-selling constraints after the suspension of all price tests in U.S. stock markets. This result speaks to the direct effect of short-selling constraints on investment, irrespective of any placebo effect.

16 The Hawthorne effect terminology refers to a number of field experiments conducted at Western Electric's Hawthorne Plant in Cicero, Illinois, in 1924. In this famous experiment, worker productivity kept on increasing whenever a change in lighting conditions was made, whether lighting conditions improved or worsened. Experimenters concluded that the act of observing workers caused the systematic increase in productivity (Landsberger 1958).

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7. Robustness

7.1 Alternate measures of financial constraints

The analysis above suggests that our results are strongest for firms that are likely to be financially constrained. But we rely on a simple definition of constraints—firms with assets below the median. While Hadlock and Pierce (2010) argue that this size definition is the strongest and simplest proxy for financial constraints, there are many other measures in the literature. In this subsection, we test whether our main results are robust to alternative measures of financial constraints.

To test whether our results are robust to alternative measures, we split the sample based on: (i) size measured by equity market capitalization, (ii) Hadlock and Pierce (2010) index of financial constraints (weighted combination of assets and firm age), (iii) a dummy variable for whether the firm has a credit rating, (iv) a dummy variable for whether the firm pays dividends, (v) a dummy variable for whether the firm has positive net payout, and (vi) the first principal component from our six measures of financial constraints (asset size, plus the five additional measures listed above).

We replicate our analysis presented in Tables 3, 4, and 6, for each of the alternative definitions of financial constraints. For each alternate definition of financial constraints, we measure the pilot group announcement return effect over the window $[-24, 11]$ as presented in the last column of Table 3, Panel B, and the two-year abnormal return difference for pilot firms as in last column of Table 4, Panel B. We then reestimate the difference-in-differences in *Capital Expenditures* for financially constrained firms in the pilot group based on the alternative measure of financial constraints.

Our results are presented in Figure 3. Overall, our results are not qualitatively sensitive to how we split the sample based on various measures of financial constraints. For example, in Panel A of Figure 3, the announcement window CARs range between -3.0% and -3.6% based on the different measures. In every case, the event returns are statistically and economically significant. There is more variation in the long-run return results, with results ranging from -1.9% to -13%. In one case, small market capitalization, our results are not statistically significant. This result is interesting because it contrasts with the strong results based on asset size. When we sort firms by size using market capitalization, small high market-to-book firms become "large" firms and large low market-to-book firms become "small" firms. Because most of the long-run underperformance in our study is coming from the small high market-to-book firms, sorting by market capitalization significantly reduces the underperformance among small firms and increases the underperformance among large firms.

Finally, Panel C of Figure 3 presents our estimates for the effect on CAPX. We find that the coefficient ranges from -0.4 to -0.9 based on our definition of financial constraints. Overall, our results do not appear to be very sensitive

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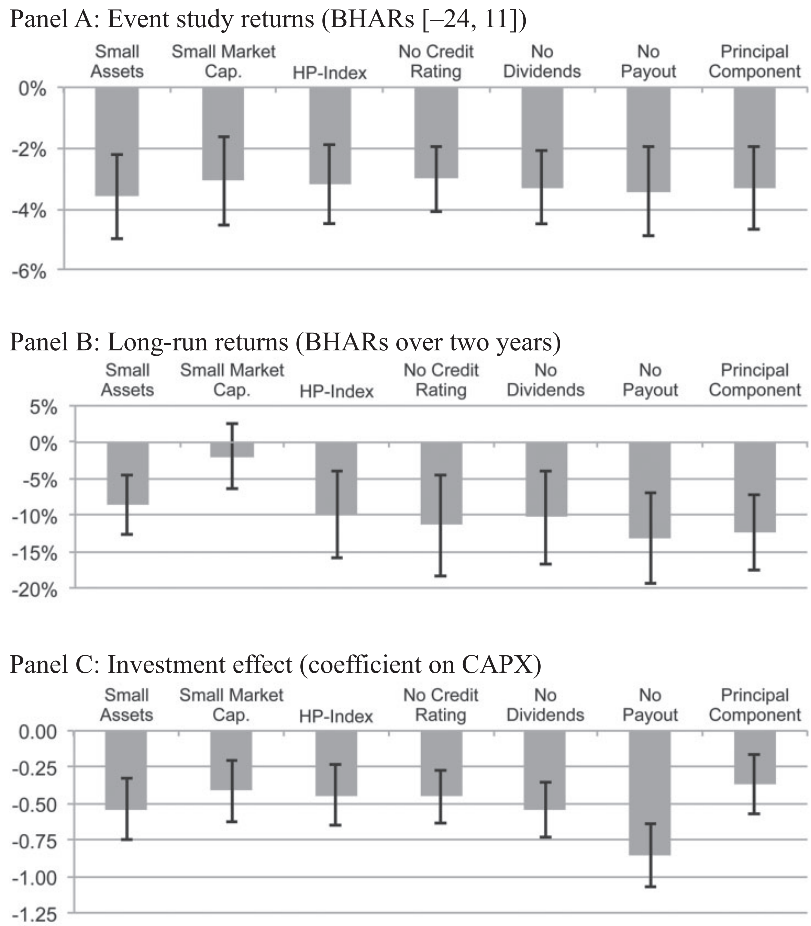

Figure 3
Sensitivity to measures of financial constraints
This graph presents a summary of our results estimated using different measures of financial constraints. Panel A presents differences in abnormal returns between the pilot and control groups in the $[-24, 11]$ (June 23 to August 12, 2004) announcement window. The daily abnormal returns are the difference between the firms' daily returns and the market's value weighted daily returns from CRSP. The reported BHARs is the difference in the buy-and-hold abnormal returns compounded daily over the event window. Panel B reports two-year market model alphas after the announcement date. Panel C reports the difference-in-differences in *Capital Expenditures* between the pilot and control group from before to after the announcement date computed as in Table 6.

to how we split the sample, and the result seems to be strongest for financially constrained pilot firms.

7.2 Monte Carlo simulations

We test whether the standard errors and coefficients that we get from our regressions could have been the result of pure chance. This is an important issue

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because Bertrand, Duflo, and Mullainathan (2004) suggest that difference-in-differences statistics may underestimate standard errors when the dependent variables are serially correlated in a panel setting. To account for this, we perform a Monte Carlo analysis in which we randomize the selection process in the pilot and the control group and perform several analyses presented earlier to test our main hypothesis. We repeat the randomization process 5,000 times and obtain empirical critical values for all of our test statistics.

Our results indicate that all t-statistics and coefficients are consistent with error rates of 5% in two-tailed tests. In particular, it should be noted that we get only three randomized samples in which we find a simultaneous reduction in *Capital Expenditures*, *Changes in Total Assets*, and *Capital Expenditures plus R&D* in the pseudo pilot groups at the 10% level. We cannot find a single randomized sample in which firms simultaneously reduce corporate spending measures and experience negative abnormal returns at the 10% level. That is, the probability that our results are driven by chance is less than about 0.02% (or 1/5,000).

In addition to these simulations, we also bootstrap an empirical distribution of results for the abnormal stock returns of firms in the pilot group. First, we randomize which firms are included in the treatment group and test whether there are abnormal returns around the twelve-day announcement window [−10, 1]. In 5,000 simulations, we find only seven cases (0.14%) of negative CARs with a t-statistic as large as we find in the data. Second, we take the actual sample of pilot firms and randomize the event window. That is, we take the true pilot stocks and form event windows based on a random start date between January 1, 2001, and June 1, 2004. In these simulations, we do not find a single example of negative returns with a t-statistic as large as we find in the data.

7.3 Trade-level analysis around the announcement date

Since we find evidence of a price effect prior to the public announcement of Reg SHO, we test whether these abnormal returns are associated with abnormal trading activity using trade-level data from TAQ. We take a standard microstructure approach to testing for differences in trading activity, trade size, and order imbalances. These results show that firms in the pilot group experience higher trading volume, larger trade sizes, and a greater proportion of seller-initiated orders than firms in the control group during the period between the SEC board approval and the public announcement date. The effects are larger for small firms. Thus, our analysis is consistent with information about the pilot group being incorporated into prices by an abnormal increase in large seller-initiated trades during this period. Informed shareholders of the pilot group seem to have sold stocks of the small firms in the pilot group early on around the Regulation SHO announcement date.

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7.4 Repeal for all stocks in 2007

On June 13, 2007, the SEC announced that the price test regulations on short sales would be repealed for all stocks effective July 6, 2007. Since pilot stocks were already under a suspended price test environment, the repeal for all stocks serves as a reverse natural experiment where the pilot stocks now serve as the control group while all other stocks receive the treatment of a price test suspension. However, unlike the announcement of the list of firms on July 28, 2004, for which there was no possibility to anticipate the effects of such an announcement, the repeal of the uptick was easier to anticipate using public information from the SEC. Therefore, this analysis is biased against finding significant results. Yet we show that even around this relatively noisy event, our main results are still there.

We follow Boehmer, Jones, and Zhang (2008), who also study the 2007 repeal. They report a significant increase in short-selling activity for the treatment group relative to the original pilot group. Consistent with their findings, we find no effect on prices in narrow windows around either the announcement or the implementation date. However, we do find some price effects for constrained stocks in a longer window from two days prior to the announcement to two days post implementation. For this sample, small control firms that had price tests repealed appear to underperform relative to the pilot group, though the statistical evidence is weaker. For these stocks, we also find slight reductions in capital expenditures and total assets (though no effect on security issuance). Again, for these results the statistical evidence is weaker. It is important to note that the implementation of the 2007 repeal happens just before an extended bear market that precedes the 2008 financial crisis. Further, the announcement of the repeal in June may not have been a surprise to some market participants, as it had already been discussed openly.[17] As a result, we must be cautious not to interpret these results too strongly. Overall prices and corporate decisions appear to move in a consistent direction around the 2007 repeal, though the economic and statistical significance are not as strong.

8. Conclusion

In this paper we test whether short-selling constraints affect stock prices and distort investment and financing decisions. We find that firms that were included in a randomly selected pilot group for the SEC's Reg SHO experienced an increase in short-selling activity and a decline in stock prices relative to a control group of stocks. Moreover, we find that the removal of short-selling constraints had a real effect on corporate decisions. Using difference-in-differences tests

[17] The SEC organized a roundtable on September 15, 2006, with finance academics who had analyzed data from the Reg SHO experiment. The transcripts suggest that the consensus was in favor of removing short-selling constraints, as most participants agreed a repeal of the uptick rule would not cause any harmful effects to financial markets.

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around the pilot program, we find that investment and equity issues declined for firms that experienced an increase in short selling. The results are stronger for small firms and for firms that were more likely to be overvalued and/or more susceptible to bear raids prior to the regulatory change.

The fact that such a seemingly small regulatory change appears to have a large effect is surprising. It is possible that our results speak to the manner in which the regulation was implemented. By focusing only on a certain subset of stocks, the SEC may have unintentionally provided a focal point for short sellers, and this may have put firms in the pilot group at a disadvantage. Thus, it is hard to determine whether our results extend to a change in short-sale regulations that affect all traded stocks. It could also be that corporate managers overreacted to being included in the pilot study simply through an increase in Knightian uncertainty caused by the unknown effects of removing a regulation that had been in place for seventy years.

Our results have a number of implications. First, we find that even a subtle regulation like the uptick rule can have a significant effect on the equity prices of financially constrained firms and appear to be a binding constraint on the equilibrium level of short selling. Second, consistent with several theoretical models, we find that the removal of short-selling constraints affects short-selling activity and stock prices even before the actual suspension of the constraints. Third, we find that corporate investment and security issues are sensitive to exogenous changes in equity prices. Our findings support both the overvaluation and bear-raids hypotheses and suggest that exogenous variation in equity prices appears to distort corporate investment flows.

Appendix

Definition of main variables

Abnormal Monthly Short Interest	The residual of a firm fixed effect regression where *Short Interest* (the monthly mean ratio of net short positions outstanding reported on the 15th of each month to shares outstanding at the start of the month) is regressed on month dummies, market-to-book, logarithm of lagged total assets, lagged return on assets, trading volume, and a dummy variable for listing on the NYSE.
CAPX	Capital expenditures (Compustat *CAPX*) scaled by start-of-year total assets (*AT*) × 100.
CAPXR&D	Capital expenditures (*CAPX*) plus research and development expenses (*XRD*) scaled by start-of-year total assets (*AT*) × 100.
Cash Flow	Net income before extraordinary items (IB) + depreciation and amortization expenses (*DP*) scaled by start-of-year total assets × 100.
Cash Holdings	Cash and short-term investment (*CHE*) scaled by start-of-year total assets (*AT*) × 100.
Δ *Total Assets*	Total assets (*AT*) divided by start-of-year total assets minus one × 100.
Debt Issues	Long-term debt issues (*DLTIS*) scaled by start-of-year total assets (*AT*) × 100.
Dividends	Common shares dividends (*DVC*) plus preferred shares dividends (*DVP*) scaled by start-of-year total assets (*AT*) × 100.
Equity Issues	Sale of common and preferred shares (*SSTK*) scaled by start-of-year total assets (*AT*) × 100.

(*continued*)

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Definition of main variables—continued

Leverage	Long-term debt (*DLTT*) plus debt in current liabilities (*DLC*) scaled by the sum of long-term debt, debt in current liabilities, and total stockholders' equity (*SEQ*) $\times$ 100.
Market-to-Book ratio	Market value of equity (*PRCC* x *CSHO*) plus book value of assets minus book value of equity minus deferred taxes (when available) (*AT-CEQ-TXDB*), scaled by book value of total assets (*AT*). Variable is lagged one year.
Monthly Short Interest	Monthly short interest reported to NASDAQ or NYSE on the 15th of each calendar month scaled by the total number of shares outstanding (from CRSP) at the start of the month.
Past Profitability	Ratio of operating income before depreciation and amortization (*OIBDP*) to start-of-year total assets (*AT*) $\times$ 100. Variable is lagged one year.
Short Interest	Average reported monthly short interest during the fiscal year, where monthly short interest reported to NASDAQ or NYSE is scaled by the total number of shares outstanding (from CRSP).
Total Assets	Start-of-year total assets (*AT*) (in million USD).
Treatment	Dummy variable equal to one if the company is in the pilot group of REG SHO and the fiscal year includes at least six months of activity after the announcement of Reg SHO (July 28, 2004) and equal to zero otherwise. The dummy variable is equal to zero for firms in the control group before the suspension of all price tests for all firms in the U.S. stock markets and equal to one after the firm's fiscal year includes at least six months after the suspension of all price tests for all firms in the U.S. stock markets (July 6, 2007).

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